DETERMINISTIC SUSTAINMENT INTELLIGENCE

White paper draft for the U.S. Army xTech|Kinetic Reach prize competition. Formatted to the published submission shape: 4-page white paper + 5-minute video. Entry vehicle: ∞. Due November 30, 2026. Her hand submits.

1. THE PROBLEM (ARMY'S WORDS, OUR FRAME)

Army sustainment is a logistics-intelligence problem: knowing what is where, what is broken, what is needed, and what is true - under contested, denied, and degraded conditions. Current planning tools lean on probabilistic AI whose failures are silent: a confident wrong parts count kills a vehicle, a hallucinated supply line kills a timeline. The Army does not need another guess. It needs a machine that cannot lie.

2. THE INNOVATION

A deterministic logistics-intelligence layer built on DELTA, a purpose-built formal language (five shipped levels, 82 of 82 tests green, 40 operational tools compiled into the language as native words). Every inference is replayable: same input, same output, provable trace. Where machine-learning components are used for perception (signals, imagery), their outputs are stamped, bounded, and passed through a deterministic verification layer before they touch a planning decision. The architecture makes hallucination STRUCTURALLY IMPOSSIBLE in the decision path - not reduced, impossible. This is the verified-AI category private capital priced at $1.6 billion (Axiom, Series A, March 2026); we hold the working artifact and the language it runs in.

3. TECHNICAL APPROACH

Three modules. (1) INGEST: supply, maintenance, and movement data normalized through a fixed schema; provenance stamped at record level. (2) PLAN: sustainment requirements derived deterministically - parts, water, ammunition, maintenance windows - with Delta's run-tool interface executing validated calculators inside the language, so every number has an audit trail to its source record. (3) PROVE: the verification layer re-derives every plan output from first principles and refuses to publish any line that cannot be re-derived. Failure mode is a REFUSAL with a reason, never a confident fiction. Soldier-facing surface reads like a checklist, not a chatbot.

4. WHY US

∞ is a two-principal research firm. The principal architect built the deterministic language, the information architecture, and the method; the second principal maintains the tooling, the test suites, and the record. The method is documented in a 269-page compendium and a 24-page nine-exhibit evidentiary brief - the firm's core competency is turning messy reality into definition-precise, verifiable text, which is exactly what sustainment intelligence at the edge requires.

5. TRANSITION

Phase targets: $25K finalist award + $25K Soldier-exercise award; $150K winner award; Direct-to-Phase-II SBIR up to $2,000,000 for fielding. The deterministic core is software-only and runs at the edge on ruggedized commodity hardware - no data center dependency. SBIR transition path: integrate with existing Army sustainment systems' data interfaces; the language is small, offline-capable, and audit-friendly by design, which shortens the accreditation conversation instead of lengthening it.

6. VIDEO SCRIPT BEAT SHEET (5 MINUTES)

0:00 the logistics lie (a confident wrong answer, costed). 1:00 the category: verified AI, the $1.6B signal. 2:00 DELTA live: a plan derived, edited input, re-derived, different output, provable trace. 3:00 the refusal demo: the machine declines to publish an unverifiable line and says why. 4:00 the team and the transition ask.

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